

Kelvin Zigah

[Email](#) | [LinkedIn Profile](#) | [GitHub](#) | [Portfolio Website](#)

Experience

Optical Hardware Intern, Fonex Data Systems Sep 2025–Present
Montreal, QC

- Developed and executed test procedures for SFP, SFP+, and QSFP optical transceivers using BERT equipment, network switches, and variable optical attenuators to validate signal integrity and compliance
- Diagnosed and resolved configuration issues on Cisco and Nokia network switches, including VLAN provisioning and MAC address management
- Identified and engaged international suppliers to expand the company's passive product portfolio with fibre Bragg grating dispersion compensation (FBG-DCM) modules
- Developed Python automation scripts to reprogram EEPROM firmware on optical transceiver modules and automate data logging into structured Excel reports

Director of Projects, IEEE Concordia Present
Concordia University, Montreal, QC

- Leading a team of 8 to design a BMS PCB for a 40A battery pack for the IEEE Ebike Project team
- Researching practical designs and applications for RISC-V CPU to get students in ECE interested in ASIC design
- Advising strategies for a magnetic encoder PCB for a 6-DOF robotic arm project led by other teams at IEEE

GLV and Firmware sub-team, Concordia Formula Electric Present
Concordia University, Montreal, QC

- Designing PCBs in Altium Designer for GLV sub-systems for the 2026 Prototype car to be tested in March.
- Learning fundamental concepts in building electronics and firmware designs for data acquisition on an electric vehicle via CAN-BUS.

Lab Supervisor, IEEE Concordia Sep 2024–Present
Concordia University, Montreal, QC

- Led circuit design and FPGA workshops, troubleshooting microcontroller and power electronics circuits with oscilloscopes and multimeters, enhancing practical automation skills
- Managed lab equipment and fostered collaboration among IEEE members for robotics projects

Projects

STM32 PCB design July 2025
Personal | Analog Circuit Design, Altium Designer, LTSpice, STM32CubeIDE

- Designed an STM32-based MCU PCB in Altium Designer with low power circuitry using industry standards and practices for EMI regulation
- Simulated power circuitry in LTSpice and prototyped with through-hole components on a perfboard
- Integrated I2C and UART busses via the STM32CubeIDE

RISC-V CPU on Altera DE2-115 FPGA Jan 2025–Present
IEEE Concordia | SystemVerilog, FPGA, Embedded Systems

- Designed a multi-cycle RISC-V CPU with RV32I ISA in SystemVerilog, integrating UART/SPI for real-time data exchange and enhancing embedded control capabilities
- Verified design with ModelSim and synthesized with Quartus, achieving optimized timing for real-time applications
- Programmed on Altera DE2-115 FPGA board, achieving responsive, real-time digital display control for practical applications, currently integrating pipelining and hazard/interrupt controls

TSSI and RTM indicator PCB design Ongoing
Concordia Formula Electric | Analog Circuit Design, Altium Designer, LTSpice

- Developing PCBs for the TSSI indicator and ready-to-move lights on the main chassis of the race car

Skills

- **Hardware:** FPGA design, Analog Circuits, Digital Design, ARM Assembly, Firmware
- **Programming:** C++, Python, SystemVerilog, Embedded C
- **Tools:** Quartus Prime, ModelSim, LTSpice, Altium Designer, Vivado, Git, Linux
- **Soft Skills:** Leadership, Teamwork, Communication

Education

Bachelor of Electrical Engineering, Co-op

Expected May 2028

Concordia University, Montreal, QC

Relevant Coursework: Computer Architecture, Digital Design, Circuit Analysis, Signals & Systems

GPA: 3.3